

Unveiling Strategic Nodes in Global Food Trade Networks: A HITS-Based Approach with Geographical Bias Correction

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1 Introduction and Research Problem

As agricultural production is unevenly distributed across the globe, the stability of the supply chain relies on a few critical players—producers who act as "authorities" and intermediaries who serve as "hubs." Understanding the structural importance of these nations is vital for assessing the resilience of the global food system against disruptions. Traditional analysis often relies on simple trade volumes, which may overlook the underlying topology and the strategic roles countries play in directing resource flows.

This research addresses these gaps by implementing an enhanced HITS (Hyperlink-Induced Topic Search) algorithm integrated with distance-decayed correction. By incorporating geographical constraints into the mutual reinforcement logic of Hubs and Authorities, this study aims to filter out localized noise and unveil the hidden strategic nodes that underpin the resilience and efficiency of the global food supply chain.

2 Dataset Description

The empirical analysis of this study relies on the integration of two distinct datasets to construct a spatially-aware trade network.

Food Trade Flow Data: The primary dataset is sourced from the FAOSTAT Detailed Trade Matrix, which provides comprehensive annual records of bilateral agricultural trade. This dataset includes granular information such as the Reporter Countries, Partner Countries, the specific food commodities traded, and the monetary Value of these transactions. For this project, we focused on the most recent complete trade cycle to capture the contemporary structure of the global food supply chain, establishing a Regional Baseline focused on the European trade network.

Geographic Coordinate Data: To account for the spatial friction in trade, we utilized the Google Canonical Country Coordinates dataset. This reference set provides the definitive latitude and longitude for every sovereign nation. By mapping these coordinates to the FAOSTAT trade matrix, we transition from a purely topological graph to a geospatial network. These coordinates serve as the foundation for calculating the Great-circle distance between trading partners.

3 Planned Work and Methodology

This project mainly relies on following works consisting of 4 stages:

- 1. Data Preparation:** This stage involves the synthesis of the FAOSTAT trade matrix and Google's geospatial coordinates. We will clean the raw CSV data to handle missing values and align inconsistent country names. The primary output is a refined dataset where each trade flow is mapped to the precise latitude and longitude of the source and destination nations, enabling the calculation of great-circle distances. Furthermore, specific trade commodities will

be selected based on their capacity to accentuate the structural roles of nodes within the global supply chain.

2. Graph Construction: We will model the food trade system as a directed, weighted graph. Unlike traditional models that use trade volume as the sole weight, we will construct a Geo-Weighted Adjacency Matrix where each edge weight is redefined using a distance-decay function to mathematically penalize "accidental" border trade and prioritize long-range strategic dependencies.

3. Network Profiling: Using the custom weights, we will implement the HITS algorithm to iteratively compute Authority and Hub scores for each nation. This step goes beyond simple statistics to profile the structural roles of countries: identifying primary producers as "Authorities" and critical resource allocators or re-exporters as "Hubs."

4. Model Evaluation: The final stage focuses on validating the impact of the geographic bias correction. We will perform a Spearman Rank Correlation analysis to compare our results against a baseline HITS model. Furthermore, we will conduct sensitivity and robustness tests to ensure that our strategic rankings remain consistent under various levels of spatial friction and data noise.

4 Algorithms and Techniques

Graph Modeling: A directed weighted graph $G = (V, E)$ will be constructed to mimic the behavior of the trade network, where V is the set of countries, E represents trade links.

HITS Algorithm: HITS will be employed to decipher the dual structural roles inherent in the trade network. We will compute authority scores (a) to identify primary production centers and hub scores (h) to distinguish strategic intermediaries or distribution centers. These scores are derived through a process of mutual reinforcement, where the values are iteratively updated until numerical convergence is achieved, thereby unveiling the hierarchy of influence within the agricultural market.

Distance-Decayed Correction: To mitigate the "proximity bias" inherent in traditional trade models, we will introduce a geospatial friction factor into the adjacency matrix. This correction would ensure that long-range strategic dependencies are prioritized over routine border trade, allowing the HITS algorithm to extract the true global influence of nodes by penalizing links that exist merely due to geographical convenience.

Distance-Decayed Correction: To mitigate the "proximity bias" inherent in traditional trade models, we will introduce a geospatial friction factor into the adjacency matrix. Let V_{ij} be the trade volume and d_{ij} be the Great-circle distance between country i and j . The corrected edge weight W_{ij} will be defined using a distance-decay function, such as:

$$W_{ij} = V_{ij} \times f(d_{ij}) \quad \text{where} \quad f(d_{ij}) = 1 - e^{-\beta d_{ij}} \quad (\text{or similar forms}) \quad (1)$$

This correction ensures that long-range strategic dependencies are prioritized over routine border trade, allowing the HITS algorithm to extract the true global influence of nodes.

5 Model Evaluation and Metrics

To ensure the validity and reliability of the geospatial framework, the model will be evaluated across three critical dimensions: computational convergence, ranking consistency, and structural robustness:

Computational Convergence: We will evaluate the computational stability of the modified HITS algorithm by monitoring the convergence of the Authority (a) and Hub (h) vectors. The iterative process is considered successful if the change in scores between successive iterations, measured by the L2-norm, falls below a predefined threshold. A rapid and stable

convergence rate will demonstrate that the introduction of the distance-decay weight does not compromise the numerical integrity of the original algorithm.

Ranking Consistency: To assess the impact of the geographic correction, we will conduct a comparative analysis between our distance-decayed model and the unweighted baseline. We utilize the Spearman Rank Correlation Coefficient (ρ) to measure the degree of similarity between the two rankings. A moderate correlation (e.g., $0.7 < \rho < 0.9$) would indicate that while the global trade hierarchy remains recognizable, our model effectively "re-ranks" nations by penalizing accidental border trade and elevating countries with long-range strategic influence.

Structural Robustness: The reliability of the model is further validated through robustness testing, which encompasses two primary evaluative approaches. First, edge perturbation is conducted by randomly removing a small percentage of trade links to observe whether the hierarchy of top-ranked nodes remains stable under network interference. Second, parameter sensitivity is addressed by testing various decay functions to ensure the final strategic insights are insensitive to the specific functional form chosen.

6 Expected Deliverables

By the end of the deadline, following materials would be presented:

Python scripts that can construct and visualize the designated FAO trade network.

An around 8-page final research report and corresponding slides, encompassing the mathematical background, network construction, quantitative evaluation, empirical results, and a discussion of the real-world economic implications based on our findings.

7 Declaration of LLM Usage

In the preparation of this proposal, the authors utilized large language models exclusively as auxiliary tools for linguistic refinement. Specifically, AI was employed to polish the academic prose, enhance the clarity of technical descriptions. The core conceptual framework remains the original work of the authors. All final content has been meticulously reviewed and validated to ensure accuracy and academic integrity.

List of LLM Used: Gemini, ChatGPT, NotebookLM

References

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